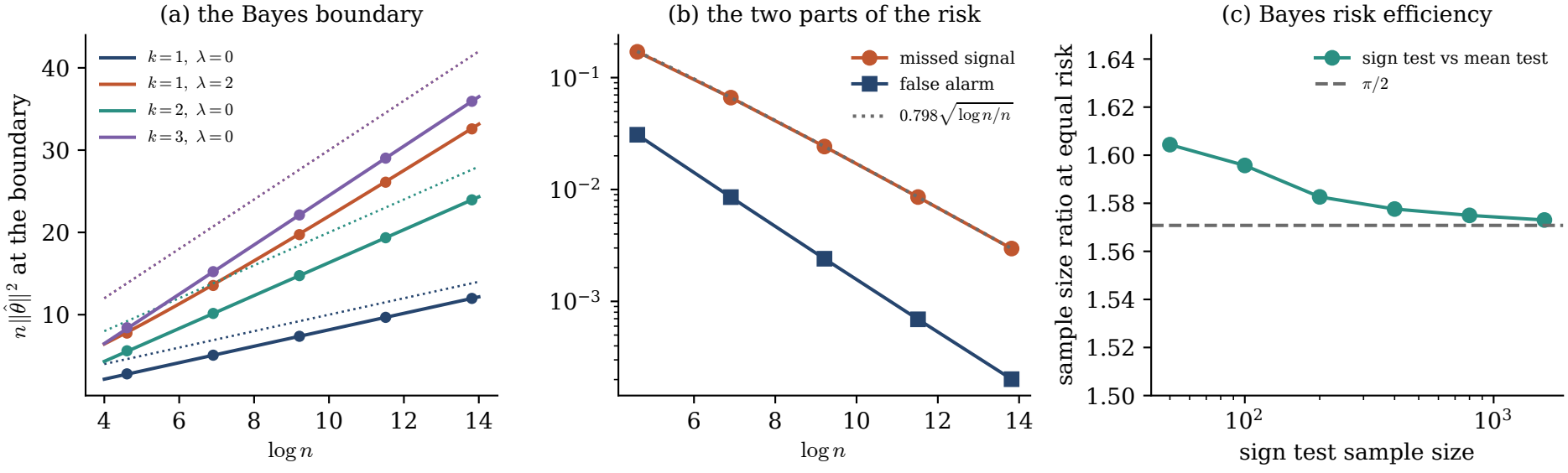
(a) the Bayes boundary
$k=1,\ \lambda=0$
$k=1,\ \lambda=2$
$k=2,\ \lambda=0$
$k=3,\ \lambda=0$
$n\|\hat{\theta}\|^2$ at the boundary
40
30
20
10
4
6
8
10
12
14
$\log n$
(b) the two parts of the risk
missed signal
false alarm
$0.798\sqrt{\log n/n}$
10^{-1}
10^{-2}
10^{-3}
6
8
10
12
14
$\log n$
(c) Bayes risk efficiency
sign test vs mean test
$\pi/2$
sample size ratio at equal risk
1.64
1.62
1.60
1.58
1.56
1.54
1.52
1.50
10^2
10^3
sign test sample size